

Project – Spotify Analysis

Introduction:

This Power BI implementation project focuses on analyzing Spotify listening history data to uncover trends in user behavior, preferred content, and listening time distribution. The dashboard provides interactive visuals that summarize top artists, albums, and tracks, while also comparing year-over-year changes and analyzing listening patterns by day type (weekday vs. weekend). Leveraging Power BI's advanced DAX capabilities and visual flexibility, this solution helps users gain deep insights into their music preferences and habits over time.

Steps:

1. Imported Data

- Loaded Spotify listening history from pre-processed CSV format.
- Verified row counts for data integrity.

2. Checked Data Quality

- Opened Power Query Editor.
 - Enabled Column Quality, Distribution, Profile.
 - Ensured:
 - 100% valid values for ts, track_name, artist_name, album_name.
 - 0% errors.
 - Transformed data:
 - Converted ts to DateTime.
 - Created calculated fields
- ```
Track Played Date = DATE(YEAR(spotify_history[ts]), MONTH(spotify_history[ts]), DAY(spotify_history[ts]))
Track Played Time = FORMAT(spotify_history[ts], "HH:MM:SS")
```

#### 3. Created Date Table

- Created a new table  

```
Date Table = CALENDAR(MIN(spotify_history[Track Played Date]), MAX(spotify_history[Track Played Date]))
```
- Added columns: Year, Day and Weekday\_Weekend.  

```
Year = YEAR('Date Table'[Date])
Day = FORMAT('Date Table'[Date], "DDD")
Weekday_Weekend =
IF(
 WEEKDAY('Date Table'[Date], 2) <= 5,
 "Weekday",
 "Weekend"
)
```
- Created relationship with spotify\_history to 'Date Table' (one-to-many).  

```
[spotify_history[Track Played Date] <— Date Table[Date]]
```

#### 4. Created Measures

- **Total Counts Measures:**  

```
No of Albums = DISTINCTCOUNT(spotify_history[album_name])
No of Artists = DISTINCTCOUNT(spotify_history[artist_name])
No of Tracks = DISTINCTCOUNT(spotify_history[track_name])
```
- **Time-Intelligent Measures:**  

```
PreviousYearAlbums =
VAR _CurrentYear = MAX('Date Table'[Year])
VAR _PreviousYear = _CurrentYear - 1
RETURN
CALCULATE(
 DISTINCTCOUNT(spotify_history[album_name]),
 'Date Table'[Year] = _PreviousYear
)
CurrentYearAlbums =
VAR _CurrentYear = MAX('Date Table'[Year])
VAR _AlbumCount =
CALCULATE(
 DISTINCTCOUNT(spotify_history[album_name]),
```

```

 'Date Table'[Year] = _CurrentYear
)
RETURN
 _AlbumCount + 0
PreviousYearArtists =
VAR _CurrentYear = MAX('Date Table'[Year])
VAR _PreviousYear = _CurrentYear - 1
RETURN
 CALCULATE(
 DISTINCTCOUNT(spotify_history[artist_name]),
 'Date Table'[Year] = _PreviousYear
)
CurrentYearArtists =
VAR _CurrentYear = MAX('Date Table'[Year])
VAR _ArtistCount =
 CALCULATE(
 DISTINCTCOUNT(spotify_history[artist_name]),
 'Date Table'[Year] = _CurrentYear
)
RETURN
 _ArtistCount + 0
PreviousYearTracks =
VAR _CurrentYear = MAX('Date Table'[Year])
VAR _PreviousYear = _CurrentYear - 1
RETURN
 CALCULATE(
 DISTINCTCOUNT(spotify_history[track_name]),
 'Date Table'[Year] = _PreviousYear
)
CurrentYearTracks =
VAR _CurrentYear = MAX('Date Table'[Year])
VAR _TrackCount =
 CALCULATE(
 DISTINCTCOUNT(spotify_history[track_name]),
 'Date Table'[Year] = _CurrentYear
)
RETURN
 _TrackCount + 0

```

- **KPI Measures:**

```

PY & YoY Albums KPI =
VAR _Current = [CurrentYearAlbums]
VAR _Previous = [PreviousYearAlbums]
VAR _YoY = IF(NOT(ISBLANK(_Previous)), DIVIDE(_Current - _Previous, _Previous, 0),
BLANK()) -- YoY %
RETURN
 IF(
 NOT(ISBLANK(_Previous)),
 "vs PY: " & FORMAT(_Previous, "#,##0") & " (" & FORMAT(_YoY, "0.00%") & ")",
 "No Data"
)
PY & YoY Artists KPI =
VAR _Current = [CurrentYearArtists]
VAR _Previous = [PreviousYearArtists]
VAR _YoY = IF(NOT(ISBLANK(_Previous)), DIVIDE(_Current - _Previous, _Previous, 0),
BLANK()) -- YoY %
RETURN
 IF(
 NOT(ISBLANK(_Previous)),
 "vs PY: " & FORMAT(_Previous, "#,##0") & " (" & FORMAT(_YoY, "0.00%") & ")",
 "No Data"
)
PY & YoY Tracks KPI =
VAR _Current = [CurrentYearTracks]
VAR _Previous = [PreviousYearTracks]
VAR _YoY = IF(NOT(ISBLANK(_Previous)), DIVIDE(_Current - _Previous, _Previous, 0),
BLANK()) -- YoY %

```

```

RETURN
 IF(
 NOT(ISBLANK(_Previous)),
 "vs PY: " & FORMAT(_Previous, "#,##0") & " (" & FORMAT(_YoY, "0.00%") & ")",
 "No Data"
)
)

```

- **MinMax Measures:**

```

MinMaxAlbums =
VAR _MaxValue = MAXX(ALLSELECTED('Date Table'[Year]),
 CALCULATE(DISTINCTCOUNT(spotify_history[album_name])))
VAR _MinValue = MINX(ALLSELECTED('Date Table'[Year]),
 CALCULATE(DISTINCTCOUNT(spotify_history[album_name])))
VAR _CurrentValue = DISTINCTCOUNT(spotify_history[album_name])
RETURN
 IF(_CurrentValue = _MaxValue || _CurrentValue = _MinValue, _CurrentValue, BLANK())

MinMaxArtists =
VAR _MaxValue = MAXX(ALLSELECTED('Date Table'[Year]),
 CALCULATE(DISTINCTCOUNT(spotify_history[artist_name])))
VAR _MinValue = MINX(ALLSELECTED('Date Table'[Year]),
 CALCULATE(DISTINCTCOUNT(spotify_history[artist_name])))
VAR _CurrentValue = DISTINCTCOUNT(spotify_history[artist_name])
RETURN
 IF(_CurrentValue = _MaxValue || _CurrentValue = _MinValue, _CurrentValue, BLANK())

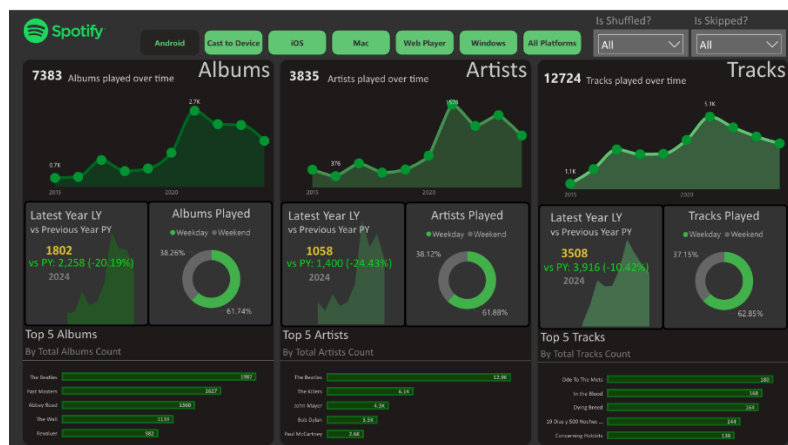
MinMaxTracks =
VAR _MaxValue = MAXX(ALLSELECTED('Date Table'[Year]),
 CALCULATE(DISTINCTCOUNT(spotify_history[track_name])))
VAR _MinValue = MINX(ALLSELECTED('Date Table'[Year]),
 CALCULATE(DISTINCTCOUNT(spotify_history[track_name])))
VAR _CurrentValue = DISTINCTCOUNT(spotify_history[track_name])
RETURN
 IF(_CurrentValue = _MaxValue || _CurrentValue = _MinValue, _CurrentValue, BLANK())

Max Year = MAX('Date Table'[Year])

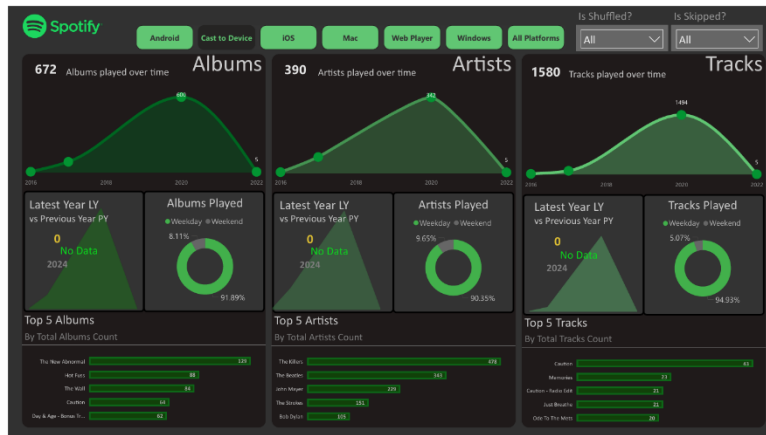
```

5. Created Dashboards with page specific filters of each Platform Type
6. Created A Final Dashboard to show insights for All Platform Types
7. Dashboard Layout and Visuals
  - Page Navigator
  - Header Slicers
    - Is Shuffled?, Is Skipped?
  - Top Row (Trend Area Charts)
    - Albums, Artists, Tracks over time using area charts
  - Middle Row (KPIs and Donuts)
    - Cards: CurrentYearAlbums, Artists, Tracks
    - Donut Charts: Listening Day Type (Weekday vs. Weekend) for Albums, Artists, Tracks
  - Bottom Row (Top 5 Bar Charts)
    - Top 5 Albums, Artists, Tracks by total listens (Top N filtering)

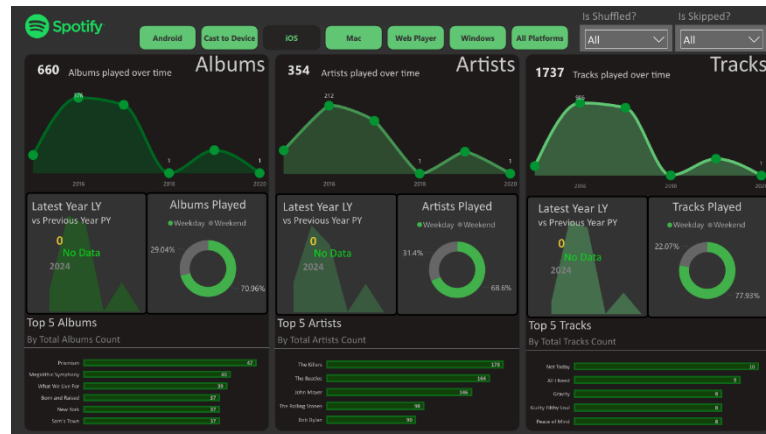
#### Dashboard for Android Platform



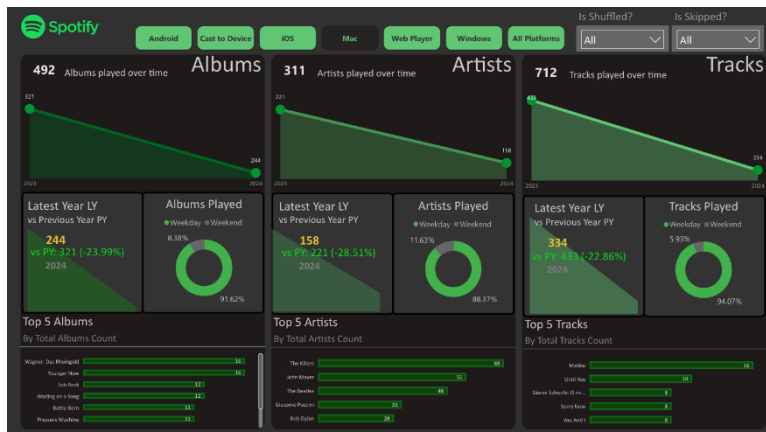
## Dashboard for Cast to Device Platform



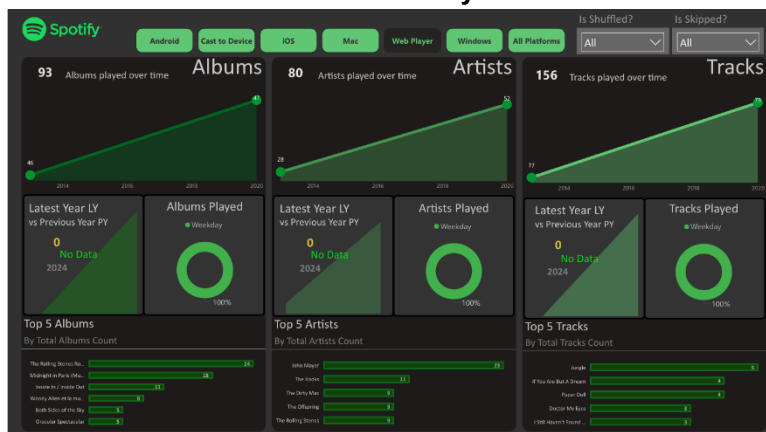
## Dashboard for iOS Platform



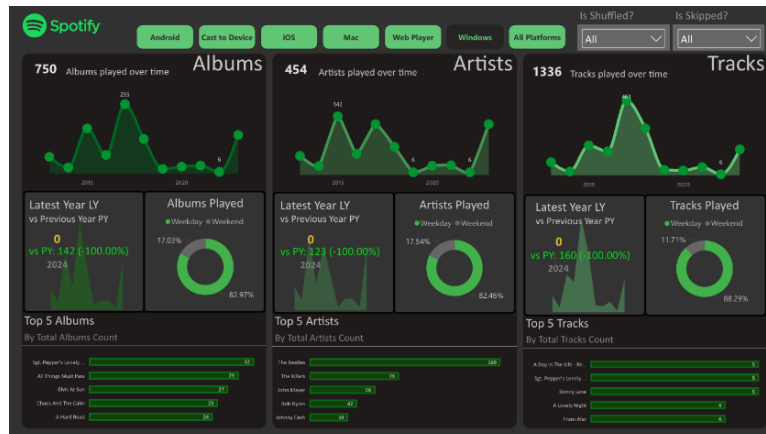
## Dashboard for Mac Platform



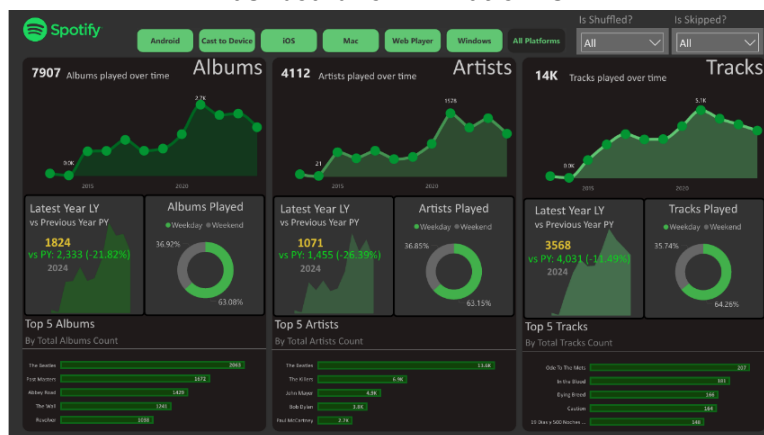
## Dashboard for Web Player Platform



## Dashboard for Windows Platform



## Dashboard for All Platforms



### 8. Dashboard Design and Formatting

- Dark theme background
- Green and white color palette
- Rounded corners and visual shadows
- Consistent font and title formatting
- Spotify logo for branding

### Key Insights:

#### 1. Listening Trends Over Time

- **Albums Played:** 7,907 total albums, with a decline in recent years (e.g., 2024 vs. previous year: -21.82% for albums, -26.39% for artists).
- **Weekday vs. Weekend Listening:**
  - **Albums:** 36.92% weekend, 63.08% weekday.
  - **Artists:** 36.85% weekend, 63.15% weekday.
  - **Tracks:** 35.74% weekend, 64.26% weekday.
- **Insight:** Weekday listening dominates, suggesting lower engagement during leisure time.

#### 2. Top Content

- **Top 5 Albums:** The Beatles (2,063 plays), Past Makeries (1,672), Abbey Road (1,429).
- **Top 5 Artists:** The Beatles (13.6K plays), The Killers (6.9K), John Mayer (4.9K).
- **Top 5 Tracks:** Oaks To The Mars (207 plays), In the Blood (181), Dying Bread (166).
- **Insight:** Classic artists like The Beatles dominate, indicating enduring popularity.

#### 3. Year-over-Year Decline

- **2024 vs. Previous Year (PY):**
  - Albums: -21.82% (1,824 vs. 2,333).

- Artists: -26.39% (1,071 vs. 1,455).
- Tracks: -11.49% (3,568 vs. 4,031).
- **Insight:** Significant drop in engagement, possibly due to changing user habits or platform shifts.
- 4. **Platform Usage**  
Android, Cast to Device, iOS, Mac, Web Player, Windows.
- 5. **Inconsistencies in Data**
  - **Some Platforms** have partial or missing data (shown as "No Data" for 2024 comparisons).
  - **Insight:** Potential gaps in tracking or reporting, limiting trend analysis.
- 6. **Niche Trends**
  - Some Platforms highlight niche albums/artists (e.g., *Wrapper Data Distributed*, *The Editor*), suggesting fragmented listening habits beyond mainstream hits.
- 7. **Overall Insights:**
  - Weekday listening drives higher engagement, ideal for targeted promotions.
  - Classic rock (e.g., The Beatles) remains a staple, but newer content struggles to match its dominance.
  - Yearly declines warrant investigation into user retention or platform competitiveness.
  - Data gaps highlight the need for improved tracking to refine insights.

### **Conclusion:**

The Spotify Analysis dashboard provides a visually engaging and data-rich summary of an individual's listening history. Through KPIs, top content lists, and time-based visuals, users can easily identify their evolving music preferences and compare engagement patterns across different years and platforms. The modular and theme-consistent design ensures a professional experience while offering opportunities for scaling into deeper genre, duration, or device-level analytics in the future.